

runs it. The boot manager is the code in the MBR. The standard FreeBSD boot manager is boot0, better known as Boot Easy. Again, 'Grub' is also commonly used in many Linux Distributions. FreeBSD uses boot manager boot0 by default. It is also known as stage zero of the booting process.

Interestingly, boot0 has a restricted size and capacity of 46 bytes. If you install any other operating systems after FreeBSD, they overwrite the MBR. In such cases, to replace the existing MBR with FreeBSD, you need to use the following command:

- `# fdisk -B -b /boot/boot0 device`

Further, there are four stages in the booting process.

Stages of the booting process

Stage one and two are parts of the same program. Yet, due to the unavailability of adequate space, they are stored separately. Importantly, they are on the same area of the disk installed together.

Usually, boot0 or other boot managers find a program to continue with the booting in a specific place on the disc. These two stages are on the first track of the boot slice. They begin with the first sector

itself. Again, they are outside the file systems. The boot manager finds these stages and starts loading and running them.

Stage 1: The first two stages are in the first track of the boot slice, starting from the first initial sector. Stage 1 segment of the disc stores information about the exact location of boot 2, which is the second stage. The particular slice with the required information knows how to find and execute boot 2.

Stage 2: This stage is quite sophisticated. It understands the technicalities of the FreeBSD file system. Thus, Stage 2 can find files from the storage swiftly. Another significant feature of this stage is a user-friendly interface it provides. The interface allows you to choose the loader or kernel. Subsequently, the kernel or loader brings a boot configuration



Stages in FreeBSD Booting

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